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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024 (Old)

Time: 03 Hours

Code: 100101 ***Subject: Basic Electrical Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) Ohms law is applicable to
(i) Semiconductors
(ii) Vacuum tubes
(iii) Carbon resistors
(iv) None of these

(b) The condition for maximum power transfer from a voltage source to load is
(i) the source impedance is half of the load impedance
(ii) the source impedance is equal to the load impedance
(iii) the source impedance is twice that of the load impedance
(iv) the source impedance and load impedance both must be zero

(c) The time constant of RL circuit is?
(i) 1/RL
(ii) R/L
(iii) RL
(iv) L/R

(d) The superposition theorem is used when the circuit contains.
(i) A single voltage source
(ii) Passive elements only
(iii) A number of voltage sources
(iv) All the above

(e) The power factor in an AC circuit is defined as the ratio of:
(i) Active power to reactive power
(ii) Apparent power to active power
(iii) Active power to apparent power
(iv) Apparent power to reactive power

(f) If 20, 30, and 50 ohms are connected in delta connection, then what are its equivalent values (in ohms) in star connection?
(i) 6, 10, 10
(ii) 6, 10, 15
(iii) 15, 15, 10
(iv) 6, 15, 15

(g) In a balanced three-phase system, the phase difference between any two phases is:
(i) 180˚
(ii) 90˚
(iii) 60˚
(iv) 120˚

(h) The main function of the commutator in a DC machine is to:
(i) Reduce losses
(ii) Convert AC to DC
(iii) Reverse current direction in armature winding
(iv) Provide insulation

(i) In a series RLC circuit, resonance occurs when:
(i) $X_L=X_C$
(ii) $X_L>X_C$
(iii) $X_L=R$
(iv) $R=X_C$

(j) Which tool is commonly used for desoldering?
(i) Multimeter
(ii) Breadboard
(iii) Soldering iron
(iv) Solder sucker

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Q.2 Solve both questions : **[14]**

(a) State and prove Maximum Power Transfer Theorem for a DC network. Calculate the maximum power for a given network.

(b) [Question text missing from original document, likely related to circuit analysis]

Q.3 Solve both questions : **[14]**

(a) State Norton's Theorem and draw the Norton's equivalent circuit.

(b) Discuss the difference between ideal and practical voltage and current source. Differentiate among real, reactive and apparent power.

Q.4 Solve both questions : **[14]**

(a) Discuss the analogy between the electrical and magnetic circuit.

(b) A resistor R ohms and capacitor 5μF are connected in series across a 200V dc supply. Calculate the value of R such that the voltage across the capacitor becomes 100V in 5 secs after the circuit is switched on.

Q.5 Solve both questions : **[14]**

(a) A coil has resistance of 7ohm and inductance of 3.18mH connected to 230V, 50Hz supply. Calculate: (i) phase angle (ii) power factor (iii) power consumed (iv) voltage drop across R and L.

(b) Describe the idea behind phasors. How do phasors depict voltages and currents that fluctuate sinusoidally?

Q.6 Solve both questions : **[14]**

(a) What is power factor? Discuss its importance in electrical systems, causes of low power factor, and methods of improvement.

(b) Describe the difference between star and delta connections.

Q.7 Solve both questions : **[14]**

(a) Explain the single-phase transformer's design and operation. Determine the EMF formula.

(b) Describe the construction and working principle of a DC generator. Derive the EMF equation.

Q.8 Solve this question : **[14]**

What are the general rules and guidelines for electrical installation in residential wiring? Include any relevant safety standards.

Q.9 Write short notes on any two of the following: **[14]**

(i) Application of KCL & KVL in DC Circuit
(ii) Hysteresis and Eddy Current Loss
(iii) Resonance
(iv) Working of DC Motor

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